HOMEWORK 6.3 · UNIT 6 · NC.8.F.1 · NC.8.F.5

Functions Day 3

Jan 28 / Jan 29 · Targets: F.1-6, F.5-1, F.5-3, F.5-4, F.1-4, F.1-1, F.5-9, F.5-8, F.1-9

Name _____ Date _____ Block _____

Remember: $f(x)$ means "the output when the input is x." $f(x)$ is the y-value. Ex: $f(x) = 2x + 1$, so $f(3) = 2(3) + 1 = 7$ → point (3, 7). Read graphs left to right: **increasing** = going up, **decreasing** = going down, **constant** = flat.

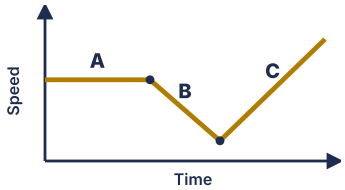
SCORE 3.0 Approaching · the basics

Targets: F.1-6, F.5-1, F.5-3, F.5-4, F.1-4

1 $f(x) = 5x + 3$. Find $f(2)$. F.1-6

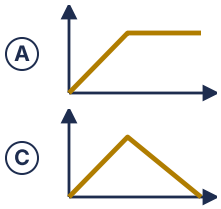
2 $f(x) = -x + 4$. Find $f(-6)$. F.1-6

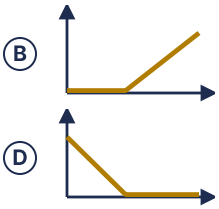
3 Which part of the graph is **increasing**? F.5-1



(A) (B) (C)

4 A kite rises at a steady rate. Then it stays at the same height. Which graph matches? F.5-3,F.5-4





5 Find the domain and range. REVIEW 6.2 F.1-4

$\{(5, -1), (7, 0), (9, -1), (11, 2)\}$

SCORE 4.0 Proficient · on target Targets: F.1-6, F.1-1

6 $f(x) = 3x + 1$. For what x is $f(x) = 16$? F.1-6

7 **Tia** made a mistake. Circle it. Explain it. Fix it. F.1-6

TIA'S WORK

$f(x) = 2x - 5$

$f(3) = 2(3) - 5 = 1$, so the point is $(1, 3)$.

8 Which is a function? (input → output) REVIEW 6.1 F.1-1

(A) a car → its license plate

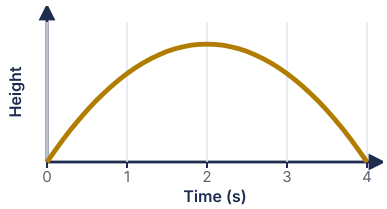
(B) a last name → people with that name

(C) a zip code → houses in it

(D) a color → things that color

SCORE 5.0 Mastery · go deeper Targets: F.5-9, F.5-8, F.1-9

9 The graph shows the height of a ball after it is thrown. When does the height increase and decrease? Is the graph linear? Explain. F.5-9,F.5-8



It increases _____ and decreases _____. It is _____ because _____.

10 Ella has \$10. Snacks cost \$2 each, so the money left is $f(x) = 10 - 2x$. Find the range for the domain $\{0, 1, 2, 3, 4, 5\}$. Does $f(6)$ make sense? REVIEW 6.2 F.1-9

The range is _____. $f(6)$ does _____ make sense because _____.